Bayesian linear Regression

Shiqi Fang

1. Motivation
2. Standard Linear Regression
3. Bayesian Linear Regression
4. Example
5. Comparison
6. When to use

Content

1. Motivation
2. Standard Linear Regression
3. Bayesian Linear Regression
4. Example
5. Comparison
6. When to use

Motivation

- Uncertainty quantification — full posterior; predictive intervals
- Prior knowledge — encode expert beliefs, especially with a small sample size
- Probabilistic predictions — event probabilities; tail-aware predictive

Content

1. Motivation
2. Standard Linear Regression
3. Bayesian Linear Regression
4. Example
5. Comparison
6. When to use

Linear Regression

Model & assumptions

$$\mathrm{y_i} = \mathrm{x_i}^\top \boldsymbol{\beta} + \varepsilon_\mathrm{i}, \quad \varepsilon_\mathrm{i} \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma^2), \qquad \mathrm{y} = \mathrm{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}, \quad \mathrm{rank}(\mathrm{X}) = \mathrm{p}.$$

OLS estimator (normal equations)

$$\hat{\boldsymbol{\beta}} = (\mathrm{X}^\top \mathrm{X})^{-1} \mathrm{X}^\top \mathrm{y} \quad \text{is the minimizer of} \quad \min_{\boldsymbol{\beta}} \|\mathrm{y} - \mathrm{X}\boldsymbol{\beta}\|_2^2.$$

Sampling distribution of $\hat{\boldsymbol{\beta}}$

- Mean derivation:

$$\mathbb{E}[\hat{\boldsymbol{\beta}}] = (\mathrm{X}^\top \mathrm{X})^{-1} \mathrm{X}^\top \mathbb{E}[\mathrm{y}] = (\mathrm{X}^\top \mathrm{X})^{-1} \mathrm{X}^\top (\mathrm{X}\boldsymbol{\beta}) = \boldsymbol{\beta}.$$

- Variance derivation:

$$\mathrm{Var}(\hat{\boldsymbol{\beta}}) = (\mathrm{X}^\top \mathrm{X})^{-1} \mathrm{X}^\top \mathrm{Var}(\mathrm{y})\, \mathrm{X} (\mathrm{X}^\top \mathrm{X})^{-1} = \sigma^2 (\mathrm{X}^\top \mathrm{X})^{-1}.$$

Thus $\hat{\boldsymbol{\beta}} \sim \mathcal{N}\big(\boldsymbol{\beta}, \sigma^2 (\mathrm{X}^\top \mathrm{X})^{-1}\big)$.

Error variance and t inference

$$\hat{\sigma}^2 = \frac{\|y - X\hat{\boldsymbol{\beta}}\|_2^2}{n-p}, \quad \frac{(n-p)\hat{\sigma}^2}{\sigma^2} \sim \chi^2_{n-p}, \quad \hat{\boldsymbol{\beta}} \perp \hat{\sigma}^2.$$

$$\Rightarrow \frac{\hat{\beta}_j - \beta_j}{\sqrt{\hat{\sigma}^2[(X^\top X)^{-1}]_{jj}}} \sim t_{n-p} \Rightarrow \hat{\beta}_j \pm t_{n-p,\,1-\alpha/2}\sqrt{\hat{\sigma}^2[(X^\top X)^{-1}]_{jj}}.$$

Prediction at a new point x_*

- Point prediction: $\hat{y}_* = x_*^\top \hat{\boldsymbol{\beta}}$.
- Mean confidence interval (no new noise):

$$x_*^\top \hat{\boldsymbol{\beta}} \pm t_{n-p,\,1-\alpha/2}\, \hat{\sigma} \sqrt{x_*^\top (X^\top X)^{-1} x_*}.$$

- Prediction interval (with new noise):

$$\hat{y}_* \pm t_{n-p,\,1-\alpha/2}\, \hat{\sigma} \sqrt{1 + x_*^\top (X^\top X)^{-1} x_*}.$$

Content

1. Motivation
2. Standard Linear Regression
3. Bayesian Linear Regression
4. Example
5. Comparison
6. When to use

Bayesian Linear Regression

Likelihood

$$\mathrm{y} \mid \mathrm{X}, \boldsymbol{\beta}, \sigma^2 \sim \mathcal{N}(\mathrm{X}\boldsymbol{\beta}, \sigma^2 \mathrm{I}_\mathrm{n}), \quad \mathrm{p}(\mathrm{y} \mid \boldsymbol{\beta}, \sigma^2) = (2\pi\sigma^2)^{-\mathrm{n}/2} \exp\Big(-\tfrac{1}{2\sigma^2} \|\mathrm{y} - \mathrm{X}\boldsymbol{\beta}\|_2^2 \Big),$$

$$\text{thus} \quad \mathrm{p}(\mathrm{y} \mid \boldsymbol{\beta}, \sigma^2) \;\propto\; (\sigma^2)^{-\mathrm{n}/2} \exp\Big(-\tfrac{1}{2\sigma^2} \|\mathrm{y} - \mathrm{X}\boldsymbol{\beta}\|_2^2 \Big).$$

Goal: infer $(\boldsymbol{\beta}, \sigma^2)$ and the predictive distribution for new x_*.

$$\mathrm{p}(\boldsymbol{\beta}, \sigma^2 \mid \mathrm{y}) \;\propto\; \mathrm{p}(\mathrm{y} \mid \boldsymbol{\beta}, \sigma^2)\, \mathrm{p}(\boldsymbol{\beta} \mid \sigma^2)\, \mathrm{p}(\sigma^2).$$

Conjugate priors: Normal–Inverse-Gamma $\Rightarrow$ closed-form posterior
assume:

$$\boldsymbol{\beta} \mid \sigma^2 \sim \mathcal{N}(\mathrm{m}_0,\ \sigma^2 \mathrm{V}_0), \qquad \sigma^2 \sim \text{Inv-Gamma}(\mathrm{a}_0, \mathrm{b}_0),$$

$$\mathrm{p}(\boldsymbol{\beta} \mid \sigma^2) \;\propto\; (\sigma^2)^{-\mathrm{p}/2} \exp\Big(-\tfrac{1}{2\sigma^2} (\boldsymbol{\beta} - \mathrm{m}_0)^\top \mathrm{V}_0^{-1} (\boldsymbol{\beta} - \mathrm{m}_0) \Big),$$

$$\mathrm{p}(\sigma^2) \;\propto\; (\sigma^2)^{-(\mathrm{a}_0+1)} \exp(-\mathrm{b}_0/\sigma^2).$$

Collect powers of σ^2

$$(\sigma^2)^{-\mathrm{n}/2} \cdot (\sigma^2)^{-\mathrm{p}/2} \cdot (\sigma^2)^{-(\mathrm{a}_0+1)} = (\sigma^2)^{-\left(\mathrm{a}_0+1+\frac{\mathrm{n}+\mathrm{p}}{2}\right)}.$$

Exponent: gather $\boldsymbol{\beta}$-quadratics

$$-\frac{1}{2\sigma^2}\Big[(\mathrm{y}-\mathrm{X}\boldsymbol{\beta})^\top(\mathrm{y}-\mathrm{X}\boldsymbol{\beta}) + (\boldsymbol{\beta}-\mathrm{m}_0)^\top \mathrm{V}_0^{-1}(\boldsymbol{\beta}-\mathrm{m}_0)\Big] - \frac{\mathrm{b}_0}{\sigma^2}.$$

Expand and define core matrices

$$(\mathrm{y}-\mathrm{X}\boldsymbol{\beta})^\top(\mathrm{y}-\mathrm{X}\boldsymbol{\beta}) = \mathrm{y}^\top\mathrm{y} - 2\mathrm{y}^\top\mathrm{X}\boldsymbol{\beta} + \boldsymbol{\beta}^\top\mathrm{X}^\top\mathrm{X}\boldsymbol{\beta},$$
$$(\boldsymbol{\beta}-\mathrm{m}_0)^\top\mathrm{V}_0^{-1}(\boldsymbol{\beta}-\mathrm{m}_0) = \boldsymbol{\beta}^\top\mathrm{V}_0^{-1}\boldsymbol{\beta} - 2\mathrm{m}_0^\top\mathrm{V}_0^{-1}\boldsymbol{\beta} + \mathrm{m}_0^\top\mathrm{V}_0^{-1}\mathrm{m}_0.$$

$$\boxed{\mathrm{Q} := \mathrm{X}^\top\mathrm{X} + \mathrm{V}_0^{-1}, \qquad \mathrm{h} := \mathrm{X}^\top\mathrm{y} + \mathrm{V}_0^{-1}\mathrm{m}_0.}$$

Then the $\boldsymbol{\beta}$-part is

$$\boldsymbol{\beta}^\top\mathrm{Q}\boldsymbol{\beta} - 2\mathrm{h}^\top\boldsymbol{\beta} + \underbrace{\mathrm{y}^\top\mathrm{y} + \mathrm{m}_0^\top\mathrm{V}_0^{-1}\mathrm{m}_0}_{\text{constant in } \boldsymbol{\beta}}.$$

Complete the square in $\boldsymbol{\beta}$

$$\boldsymbol{\beta}^\top \mathrm{Q}\boldsymbol{\beta} - 2\mathrm{h}^\top\boldsymbol{\beta} = (\boldsymbol{\beta} - \mathrm{m_n})^\top \mathrm{Q}(\boldsymbol{\beta} - \mathrm{m_n}) - \mathrm{m_n}^\top \mathrm{Q}\mathrm{m_n},$$

Posterior hyperparameters:

$$\boxed{\mathrm{V_n} := \mathrm{Q}^{-1} = (\mathrm{V_0}^{-1} + \mathrm{X}^\top\mathrm{X})^{-1}, \qquad \mathrm{m_n} := \mathrm{V_n}\,\mathrm{h} = \mathrm{V_n}(\mathrm{V_0}^{-1}\mathrm{m_0} + \mathrm{X}^\top\mathrm{y}).}$$

Rewrite as

$$-\frac{1}{2\sigma^2}\Big[(\boldsymbol{\beta} - \mathrm{m_n})^\top \mathrm{Q}(\boldsymbol{\beta} - \mathrm{m_n}) + \mathrm{y}^\top\mathrm{y} + \mathrm{m_0}^\top\mathrm{V_0}^{-1}\mathrm{m_0} - \mathrm{m_n}^\top\mathrm{Q}\mathrm{m_n}\Big] - \frac{\mathrm{b_0}}{\sigma^2}.$$

Posterior hyperparameters

$$\boxed{\mathrm{a_n} := \mathrm{a_0} + \tfrac{\mathrm{n}}{2}, \qquad \mathrm{b_n} := \mathrm{b_0} + \tfrac{1}{2}\Big(\mathrm{y}^\top\mathrm{y} + \mathrm{m_0}^\top\mathrm{V_0}^{-1}\mathrm{m_0} - \mathrm{m_n}^\top\mathrm{V_n}^{-1}\mathrm{m_n}\Big).}$$

Joint posterior

$$\mathrm{p}(\boldsymbol{\beta}, \sigma^2 \mid \mathrm{y}) \;\propto\; (\sigma^2)^{-\left(\mathrm{a_n}+1+\frac{\mathrm{p}}{2}\right)} \exp\Big(-\tfrac{1}{2\sigma^2}(\boldsymbol{\beta}-\mathrm{m_n})^\top \mathrm{V_n}^{-1}(\boldsymbol{\beta}-\mathrm{m_n})\Big)\, \exp\Big(-\tfrac{\mathrm{b_n}}{\sigma^2}\Big).$$

Result (closed-form posterior)

$$\boxed{\boldsymbol{\beta} \mid \sigma^2, \mathrm{y} \sim \mathcal{N}(\mathrm{m_n},\ \sigma^2 \mathrm{V_n}), \qquad \sigma^2 \mid \mathrm{y} \sim \text{Inv-Gamma}(\mathrm{a_n}, \mathrm{b_n}).}$$

Posterior predictive $\mathrm{p}(\mathrm{y}_* \mid \mathrm{x}_*, \mathrm{y})$
First integrate out $\boldsymbol{\beta}$ (Gaussian closure):

$$\mathrm{y}_* \mid \mathrm{x}_*, \sigma^2, \mathrm{y} \sim \mathcal{N}\Big(\mu_*,\ \sigma^2 \mathrm{c}_*\Big),\ \mu_* = \mathrm{x}_*^\top \mathrm{m}_\mathrm{n},\quad \mathrm{c}_* = 1 + \mathrm{x}_*^\top \mathrm{V}_\mathrm{n} \mathrm{x}_*.$$

Delete 1 in c_* if only care about the mean.

Then mix over $\sigma^2 \sim \mathrm{Inv-Gamma}(\mathrm{a}_\mathrm{n}, \mathrm{b}_\mathrm{n})$:

$$\boxed{\mathrm{y}_* \mid \mathrm{x}_*, \mathrm{y} \ \sim\ \text{Student-t}_{2\mathrm{a}_\mathrm{n}}\Big(\ \mathrm{x}_*^\top \mathrm{m}_\mathrm{n},\quad \frac{\mathrm{b}_\mathrm{n}}{\mathrm{a}_\mathrm{n}}\, \mathrm{c}_*\Big).}$$

Includes both parameter and noise uncertainty (heavier tails than Gaussian).

Content

1. Motivation
2. Standard Linear Regression
3. Bayesian Linear Regression
4. Example
5. Comparison
6. When to use

Given (from posterior):

$$x_* = \begin{bmatrix}1\\2\end{bmatrix},\ m_n = \begin{bmatrix}0.5\\1.0\end{bmatrix},\ V_n = \begin{bmatrix}0.01 & 0\\0 & 0.04\end{bmatrix},\ a_n = 150,\ b_n = 120.$$

Hence $\dfrac{b_n}{a_n} = 0.8,\ \nu = 2a_n = 300.$

Predictive mean & variance scale:

$$\mu_* = x_*^\top m_n = 0.5+2\times1.0 = 2.5, \quad x_*^\top V_n x_* = 0.01\times1^2+0.04\times2^2 = 0.17.$$

Scale for future observation:

$$\frac{b_n}{a_n}(1+0.17) = 0.8\times1.17 = 0.936 \ \Rightarrow\ \text{sd} \approx \sqrt{0.936} = 0.968.$$

95% predictive interval for future y_* (df=300, $t_{0.975}\approx1.97$):

$$2.5 \ \pm\ 1.97\times0.968 \ \approx\ 2.5\pm1.91 \ \Rightarrow\ [0.59,\ 4.41].$$

Mean response (no new noise):

$$\text{scale} = \frac{b_n}{a_n}x_*^\top V_n x_* = 0.8\times0.17 = 0.136, \quad \text{sd} \approx \sqrt{0.136} = 0.369,$$

$$95\%\ \text{t-interval:}\ 2.5\pm1.97\times0.369 \approx 2.5\pm0.73 \Rightarrow [1.77,\ 3.23].$$

When Does Predictive Uncertainty Grow?

- x_* far from the training cloud $\Rightarrow$ $x_*^\top V_n x_*$ increases.
- Multicollinearity / small n: the posterior covariance V_n inflates.
- Noisy data: larger $\frac{b_n}{a_n}$ (posterior mean of σ^2) increases overall scale.

Content

1. Motivation
2. Standard Linear Regression
3. Bayesian Linear Regression
4. Example
5. Comparison
6. When to use

Frequentist vs Bayesian

Table: Frequentist vs Bayesian (phrases)

Aspect	Frequentist	Bayesian
Parameter view	Fixed unknowns	Random variables (posterior)
Probability statements	Repeated-sampling behavior	Given-data probabilities
Point prediction	$\hat{y}_* = x_*^\top \hat{\beta}$	$\mathbb{E}[y_* \mid x_*, y] = x_*^\top m_n$
Mean-response interval	$x_*^\top \hat{\beta} \pm t\, \hat{\sigma} \sqrt{x_*^\top (X^\top X)^{-1} x_*}$	$x_*^\top \beta \mid y \sim t_{2a_n}\big(x_*^\top m_n, \frac{b_n}{a_n} x_*^\top V_n x_*\big)$
Prediction interval	$\hat{y}_* \pm t\, \hat{\sigma} \sqrt{1 + x_*^\top (X^\top X)^{-1} x_*}$	$y_* \mid x_*, y \sim t_{2a_n}\big(x_*^\top m_n, \frac{b_n}{a_n}(1 + x_*^\top V_n x_*)\big)$
Tail behavior	Gaussian	Student-t (heavier tails)

BLR Posterior Mean as a Weighted Average

Posterior mean:

$$\mathrm{m_n} \;=\; \mathrm{V_n}\big(\mathrm{V_0^{-1}m_0} + \mathrm{X^\top y}\big), \qquad \mathrm{V_n} \;=\; (\mathrm{V_0^{-1}} + \mathrm{X^\top X})^{-1}.$$

Let $\mathrm{S} = \mathrm{X^\top X}$ (> 0) and the OLS estimator $\hat{\beta}_{\mathrm{OLS}} = \mathrm{S^{-1}X^\top y}$. Then

$$\begin{aligned}\mathrm{m_n} &= (\mathrm{V_0^{-1}} + \mathrm{S})^{-1}\big(\mathrm{V_0^{-1}m_0} + \mathrm{S}\,\hat{\beta}_{\mathrm{OLS}}\big)\\ &= \frac{\mathrm{S}}{\mathrm{V_0^{-1}} + \mathrm{S}}\hat{\beta}_{\mathrm{OLS}} \;+\; \frac{\mathrm{V_0^{-1}}}{\mathrm{V_0^{-1}} + \mathrm{S}}\,\mathrm{m_0}.\end{aligned}$$

Since $\mathrm{S} > 0$ and $\mathrm{V_0^{-1}} \geq 0$, we have $\mathrm{w} \in (0, 1)$.

$$\boxed{\mathrm{m_n} \;=\; \mathrm{W}\,\hat{\beta}_{\mathrm{OLS}} \;+\; (\mathrm{I} - \mathrm{W})\,\mathrm{m_0}, \quad \mathrm{W} := (\mathrm{V_0^{-1}} + \mathrm{S})^{-1}\mathrm{S}\;.}$$

Intuition: Data Information vs Prior Precision

OLS does not care about the prior, but BLR injects the domain knowledge into the prior.

- $S = X^\top X$: data information (larger $S \Rightarrow$ stronger data evidence).
- V_0^{-1}: prior precision (larger $V_0^{-1} \Rightarrow$ stronger shrinkage toward m_0).
- Weak prior / strong data

$$V_0^{-1} \ll S \quad \Longrightarrow \quad w \approx 1 \quad \Longrightarrow \quad m_n \approx \hat{\beta}_{\mathrm{OLS}}.$$

- Strong prior / weak data

$$V_0^{-1} \gg S \quad \Longrightarrow \quad w \approx 0 \quad \Longrightarrow \quad m_n \approx m_0.$$

Hyperparameters & Sample Size: Effects on Posterior & Prediction

Param.	Role (in NIG BLR)	Posterior impact when increased	Predictive impact when increased
a_0	Inv-Gamma shape of σ^2 prior	Increases df: $a_n = a_0 + \frac{n}{2}$; less uncertainty in σ^2	Larger df $\Rightarrow$ t_ν closer to Gaussian
b_0	Inv-Gamma scale of σ^2 prior	Lifts/lowers posterior scale: b_n; affects $\mathbb{E}[\sigma^2 \mid y] \approx b_n/(a_n - 1)$	Larger $b_0 \Rightarrow$ larger scale b_n/a_n $\Rightarrow$ wider predictive bands
m_0	Prior mean of β	Pulls $m_n = (V_0^{-1} + X^\top X)^{-1}(V_0^{-1} m_0 + X^\top y)$ toward m_0	Predictions $x_*^\top m_n$ shift toward prior beliefs
λ	Prior precision on β	Via $V_0^{-1} = \lambda I$: increases shrinkage; stabilizes under collinearity	Stronger shrinkage $\Rightarrow$ mean-response bands often tighter (esp. weak-data)
n	Sample size	$X^\top X$ grows; m_n closer to OLS; V_n shrinks; a_n increases	Larger n $\Rightarrow$ predictive t df up (closer to Gaussian), bands narrower; parameter uncertainty diminishes

Notes: Mean-response variance $\propto (b_n/a_n)\, x_*^\top V_n x_*$; prediction variance adds $+\,(b_n/a_n)$.

Data Generation

$$\mathrm{n}: \text{ sample size}$$

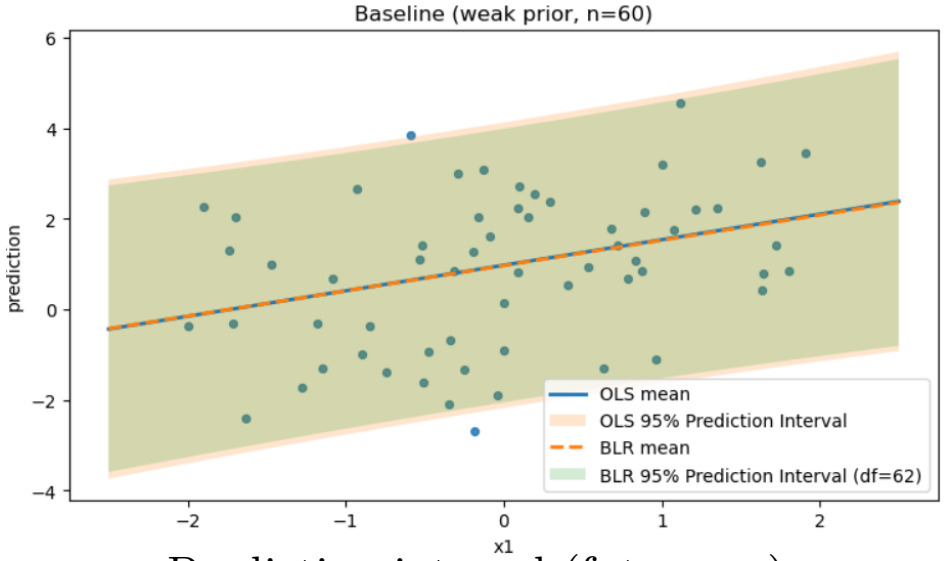
$$\mathrm{x}_1 \sim \mathrm{Unif}[-2, 2], \qquad \mathrm{x}_2 = \mathrm{x}_1 + \varepsilon_{\mathrm{c}},\ \varepsilon_{\mathrm{c}} \sim \mathcal{N}(0,\ \mathrm{sd}^2)$$

$$\mathrm{x}_3 \sim \mathrm{Unif}[-1, 1], \qquad \mathrm{X} = [1,\ \mathrm{x}_1,\ \mathrm{x}_2,\ \mathrm{x}_3] \in \mathbb{R}^{\mathrm{n}\times 4}$$

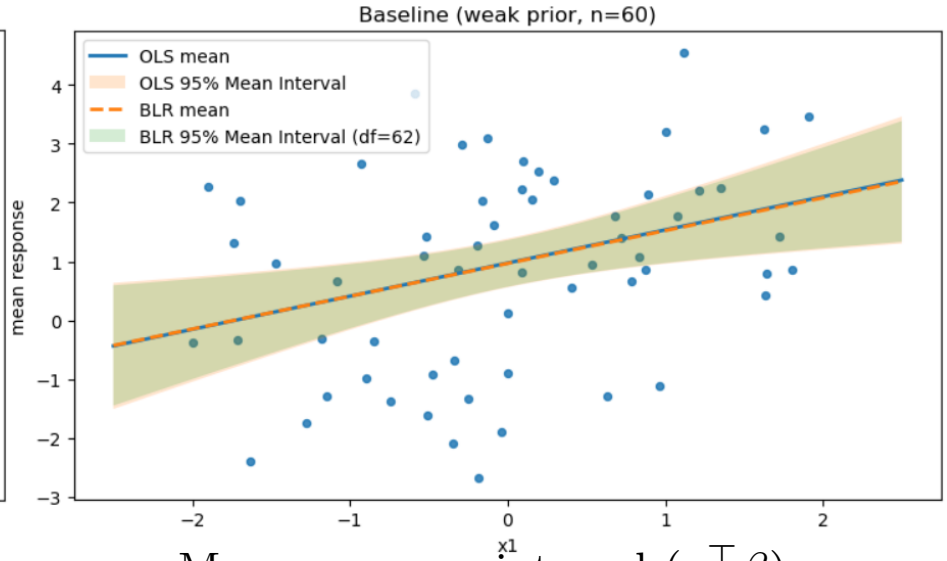
$$\beta_{\mathrm{true}} = (1.0,\ 2.0,\ -1.5,\ 0.8)^\top, \quad \sigma_{\mathrm{true}} = 1.5$$

$$\mathrm{y} = \mathrm{X}\beta_{\mathrm{true}} + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0,\ \sigma^2_{\mathrm{true}}\mathrm{I})$$

Weak Prior



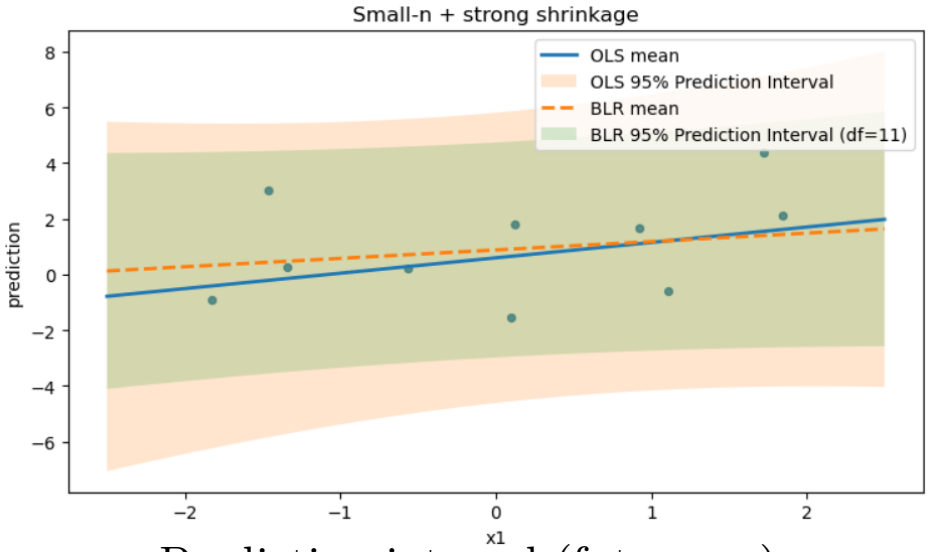
Prediction interval (future y_*)



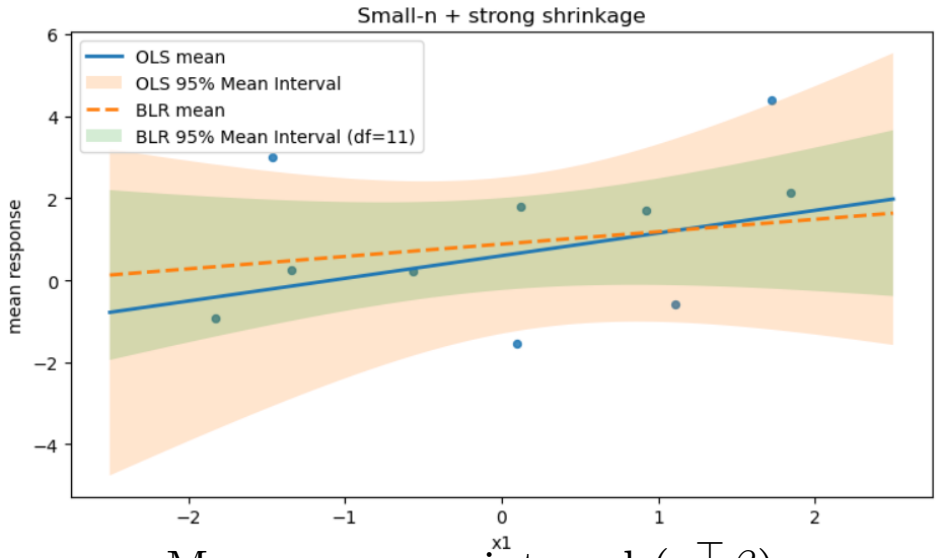
Mean-response interval ($x_*^\top \beta$)

- Settings: $n = 60$, sd$= 0.20$, $\lambda = 1.0$, scale_intercept$= 0.05$, $a_0 = b_0 = 1.0$, $m_0 = 0$.
- Neutral prior mean $m_0 = 0$; large V_0 ($V_0^{-1} = \lambda I_p$) $\Rightarrow$ weak shrinkage.
- BLR $\approx$ OLS (data-dominated); differences are minimal with a weak prior and moderate sample size n.

Strong Shrinkage



Prediction interval (future y_*)



Mean-response interval ($x_*^\top \beta$)

- Settings: $n = 10$, sd$= 0.20$, $\lambda = 25.0$, scale_intercept$= 0.05$, $a_0 = b_0 = 0.5$, $m_0 = 0$.
- Small V_0 (large V_0^{-1}) $\Rightarrow$ strong shrinkage.
- Low information (n small) $\Rightarrow$ posterior mean m_n pulled toward m_0; intervals reflect stronger prior influence.
- BLR shows tighter, more stable mean-response bands than OLS under strong shrinkage / small sample size.

Content

1. Motivation
2. Standard Linear Regression
3. Bayesian Linear Regression
4. Example
5. Comparison
6. When to use

When to use BLR

- Need probabilities and intervals, not just point estimates
- Prior knowledge available; decision/risk constraints explicit
- Small sample size, high collinearity, heavy tail scenarios